Lecture 19

Lorentzian geometry and general relativity

Objectives. *A Lorentzian metric specifies elapsed time and the paths of freely falling particles. Einstein's equation relates its curvature to matter. We recover Newton's acceleration in a weak field and verify the Schwarzschild vacuum metric. Its conserved quantities then reduce a massive particle's motion to an orbit equation.*

By the end of this lecture you will be able to:

- Distinguish timelike, null and spacelike tangent vectors in signature $+---$.
- Recover Newton's equation from a weak static metric and slow motion.
- Explain the action and conservation statements in Einstein's equation.
- Verify Schwarzschild vacuum curvature and derive the massive orbit equation.

19.1 Lorentzian metrics and proper time

A *Lorentzian spacetime* is a smooth four-dimensional manifold with a metric of signature $(+,-,-,-)$. At each point an orthonormal frame identifies the metric with $\eta = \mathrm{diag}(1,-1,-1,-1)$. This recalls the flat interval in *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Here the metric may vary from point to point. Assume the spacetime is oriented, so its metric volume form is defined.

A nonzero tangent vector v is *timelike* if $g(v,v) > 0$, *null* if $g(v,v) = 0$, and *spacelike* if $g(v,v) < 0$. The null vectors form the *light cone*. A *time orientation* is a continuous choice of one of the two timelike cones as future. We assume such a choice. In an orthonormal frame adapted to this choice, future vectors have positive time component. The figure shows one space direction and the time direction.

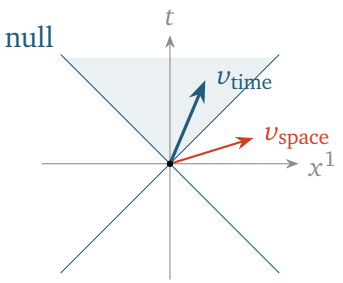


Figure 19.1: The null lines bound the future timelike cone in a tangent plane. The classification uses the metric at the point, rather than the drawing's Euclidean angles.

Use $c = 1$ and coordinates $x^0 = t$. The *proper time* along a future timelike curve $x(\sigma)$ is defined by

$$\tau(\sigma_1) - \tau(\sigma_0) = \int_{\sigma_0}^{\sigma_1} \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}\, d\sigma.$$

Its *four-velocity* $U = dx/d\tau$ satisfies $g(U, U) = 1$. The physical free-fall postulate says that an unforced test particle follows a timelike geodesic of the Levi-Civita connection. A light ray follows a null geodesic, using an affine parameter in place of proper time. These are physical inputs. The existence and equation of these geodesics are the mathematical results of Lecture 16.

19.2 Newton's equation from a weak metric

Let $\Phi = \epsilon\varphi(x^1, x^2, x^3)$ be a smooth static potential, where ϵ is a small parameter. Derivatives of Φ count as first order in ϵ. In spatial Cartesian coordinates consider

$$ds^2 = (1 + 2\Phi)dt^2 - (1 - 2\Phi) \sum_{i=1}^{3} (dx^i)^2.$$

Assume $|\Phi| < 1/2$, so the signature is Lorentzian. Define the Euclidean spatial gradient by

$$\nabla_E \Phi = (\partial_1 \Phi, \partial_2 \Phi, \partial_3 \Phi).$$

The figure shows a trajectory in its leading Newtonian approximation.

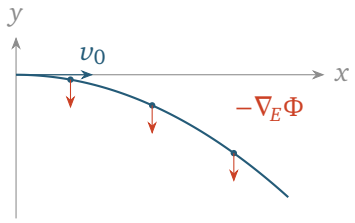


Figure 19.2: For a locally uniform downward acceleration, the leading Newtonian trajectory is $x = v_0 t$, $y = -at^2/2$, with $v_0 > 0$ and $a > 0$. This curve represents the weak-field, slow-motion limit.

Proposition 19.1 (Weak-field free fall · *proved*)**.** *For a future timelike geodesic with small coordinate speed, the coordinate-time acceleration to first order in the potential and lowest order in speed is*

$$\frac{d^2 x^i}{dt^2} = -\partial_i \Phi, \qquad i = 1, 2, 3.$$

Proof. Write $v^\mu = dx^\mu/dt$, so $v^0 = 1$. Changing from an affine parameter to t in the geodesic equation gives

$$\frac{d^2 x^i}{dt^2} = -\Gamma^i_{\mu\nu} v^\mu v^\nu + v^i \Gamma^0_{\mu\nu} v^\mu v^\nu.$$

A nonzero timelike tangent has nonzero time component in this metric, since a tangent with zero time component has negative norm. Thus division by the square of that component is valid. The affine time equation eliminates the second derivative of t when the spatial equation is divided by the square of its first derivative. The Christoffel formula gives

$$\Gamma^i_{00} = \frac{\partial_i \Phi}{1 - 2\Phi} = \partial_i \Phi + O(\epsilon^2), \qquad \Gamma^0_{0i} = \frac{\partial_i \Phi}{1 + 2\Phi}.$$

The static diagonal metric has $\Gamma^0_{00} = \Gamma^0_{ij} = \Gamma^i_{0j} = 0$. Write $\mathbf{v} = (v^1, v^2, v^3)$ for the coordinate spatial velocity. All remaining terms in the spatial equation contain at least two spatial velocity factors. The acceleration is $-\partial_i \Phi + O(\epsilon^2) + O(\epsilon |\mathbf{v}|^2)$, which proves the stated approximation. ∎

For an attractive point source of mass $m > 0$, $\Phi = -G_N m / r$, where $G_N > 0$ is Newton's constant. Its negative gradient points toward the source.

19.3 Einstein's equation and its conservation identity

Recall Lecture 17's definitions $\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$ and $R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}$. The *Einstein tensor* is

$$G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu}.$$

Take action integrals on a relatively compact coordinate domain. All variations vanish near its boundary. Let S_{m} be a matter action and let $\mu = \sqrt{|\det g|}\,\mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$ in positive coordinates. Its symmetric *stress–energy tensor* $T_{\mu\nu}$ is defined by the response to compactly supported inverse-metric variations, with matter fixed:

$$\delta S_{\mathrm{m}} = \frac{1}{2} \int T_{\alpha\beta}\, \delta g^{\alpha\beta}\, \mu.$$

The physical normalization gives $T_{00} = \rho$ for matter at rest in an orthonormal frame, where ρ is energy density. For pressureless matter of rest density ρ and unit four-velocity U, the model is $T_{\mu\nu} = \rho U_\mu U_\nu$. Write $F_{\mu\nu}$ for the components of F_{em} in this paragraph. For the field of Lecture 15, define the normalized Maxwell action by

$$S_{\mathrm{em}} = -\frac{1}{2} \int F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} = -\frac{1}{4} \int F_{\rho\sigma} F^{\rho\sigma}\, \mu.$$

This is Nakahara's (7.210b) in the present units. Vary the inverse metric while fixing the potential, hence the covariant field components. The inverse-metric factors and the determinant variation give

$$\delta(F_{\rho\sigma}F^{\rho\sigma}) = 2 F_{\alpha\lambda} F_{\beta}{}^{\lambda}\, \delta g^{\alpha\beta}, \qquad \delta\mu = -\frac{1}{2} g_{\alpha\beta}\, \delta g^{\alpha\beta}\, \mu.$$

Substitution in the action and comparison with the stress definition give

$$T_{\mu\nu} = -F_{\mu\lambda} F_{\nu}{}^{\lambda} + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

In an orthonormal frame, $F_{0\lambda} F_0{}^{\lambda} = -|\mathbf{E}|^2$ and $F_{\alpha\beta}F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$. Hence $T_{00} = (|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$ is nonnegative.

Theorem 19.2 (Einstein's equation · *cited variation; algebraic consequence proved*)**.** *The Einstein–Hilbert action in these conventions is*

$$S_{\mathrm{grav}} = -\frac{1}{16\pi G_N} \int R_{\mathrm{sc}}\, \mu.$$

For compactly supported metric variations, the cited variation formula is

$$\delta \int R_{\mathrm{sc}}\, \mu = \int G_{\alpha\beta}\, \delta g^{\alpha\beta}\, \mu.$$

Stationarity of $S_{\mathrm{grav}} + S_{\mathrm{m}}$ therefore gives

$$\mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

The variation formula is Nakahara, §7.10.2, equations (7.203)–(7.214), translated to our signature and matter convention. It discards a divergence because the variation has compact support. The displayed coefficient follows by setting $-G_{\mu\nu}/(16\pi G_N) + T_{\mu\nu}/2 = 0$. These equations are a physical field law. The action supplies their variational form.

The signature translation is explicit. Replacing a metric by its negative leaves its Christoffels, curvature operator and Ricci tensor unchanged. The scalar curvature reverses sign, so the Einstein tensor is unchanged. Nakahara's $-+++$ Einstein–Hilbert sign consequently becomes the minus sign above. For stress, his (7.211) varies the lower metric. Converting to his inverse metric contributes a minus sign by (7.203). Our inverse metric is its negative, giving a second minus sign. The two cancel, yielding our positive inverse-metric variation with the same lowered stress tensor.[1] Carroll's free notes use the same curvature index order as ours.[2]

For the weak metric above, the additional first-order symbols are

$$\Gamma^i_{jk} = -\delta^i_k \partial_j \Phi - \delta^i_j \partial_k \Phi + \delta_{jk} \partial_i \Phi, \qquad \Gamma^\mu_{\mu i} = -2\partial_i \Phi.$$

Here and in the next calculation terms of order ϵ^2 are omitted. The quadratic Christoffel terms do not contribute at first order. Staticity removes time derivatives and gives $\mathrm{Ric}_{0i} = 0$. Thus $\mathrm{Ric}_{ij} = \partial_k \Gamma^k_{ji} - \partial_j \Gamma^\mu_{\mu i}$: the mixed second derivatives cancel, leaving $\delta_{ij} \sum_k \partial_k^2 \Phi$. The full contraction gives

$$\mathrm{Ric}_{00} = \nabla_E^2 \Phi, \qquad \mathrm{Ric}_{ij} = \delta_{ij} \nabla_E^2 \Phi, \qquad R_{\mathrm{sc}} = -2\nabla_E^2 \Phi, \qquad G_{00} = 2\nabla_E^2 \Phi.$$

Here $\nabla_E^2 = \sum_i \partial_i^2$ is the Euclidean spatial Laplacian. For example, at this order $\mathrm{Ric}_{00} = \partial_i \Gamma^i_{00}$. The 00 field equation for slowly moving matter gives $\nabla_E^2 \Phi = 4\pi G_N \rho$, the positive Newtonian Poisson equation. This fixes the Einstein sign independently of a source's notation.

The conservation identity follows from the covariant second Bianchi identity of Proposition 17.4, whose conversion from forms was sketched. Here is its full contraction. Use the lowered tensor $R_{ab\mu\nu}$ of Lecture 17. Metric compatibility and the two pair antisymmetries give

$$\begin{aligned} 0 &= g^{a\mu} g^{b\lambda} \left(\nabla_\lambda R_{ab\mu\nu} + \nabla_\mu R_{ab\nu\lambda} + \nabla_\nu R_{ab\lambda\mu} \right) \\ &= 2\nabla^\mu \mathrm{Ric}_{\mu\nu} - \partial_\nu R_{\mathrm{sc}}. \end{aligned}$$

The three contractions are respectively $\nabla^\mu \mathrm{Ric}_{\mu\nu}$, $\nabla^\mu \mathrm{Ric}_{\mu\nu}$ and $-\partial_\nu R_{\mathrm{sc}}$. Thus $\nabla^\mu G_{\mu\nu} = 0$. Einstein's equation implies $\nabla^\mu T_{\mu\nu} = 0$. This is a local covariant conservation statement. It alone does not assert a globally conserved energy on an arbitrary spacetime.

19.4 The Schwarzschild vacuum metric

Now use geometric units $G_N = c = 1$. Let $M > 0$ denote the mass parameter, equal to $G_N m/c^2$ before this unit choice. On the exterior $r > 2M$, set $f(r) = 1 - 2M/r$. The *Schwarzschild*

[1] Use (7.211) for this translation. The inverse-metric prefactor in (7.212) needs a minus sign to agree with it.

[2] Carroll, *Lecture Notes on General Relativity* (1997), (3.67), (3.90)–(3.96), pp. 76, 81–82. His metric has the opposite sign. His Schwarzschild geodesic formulas (7.33)–(7.48), pp. 172–174, translate by this metric reversal.

metric is

$$\mathrm{d}s^2 = f\,\mathrm{d}t^2 - f^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

The angular expression is the round-sphere metric, with usual overlapping charts at its coordinate poles.

Proposition 19.3 (Schwarzschild vacuum · *proved with exact algebra check*)**.** *The exterior Schwarzschild metric has* $\mathrm{Ric} = 0$*. The fields* ∂_t *and* ∂_ϕ *are Killing fields. Along an affinely parametrized geodesic, the quantities*

$$E = f\dot{t}, \qquad L = r^2\sin^2\theta\,\dot{\phi}$$

are constant.

Proof. First compute for any smooth positive function $f(r)$. Up to symmetry in the two lower indices, the nonzero Christoffels are

$$\begin{array}{lll} \Gamma^t_{tr} = f'/(2f), & \Gamma^r_{tt} = ff'/2, & \Gamma^r_{rr} = -f'/(2f), \\ \Gamma^r_{\theta\theta} = -rf, & \Gamma^r_{\phi\phi} = -rf\sin^2\theta, & \Gamma^\theta_{r\theta} = 1/r, \\ \Gamma^\phi_{r\phi} = 1/r, & \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, & \Gamma^\phi_{\theta\phi} = \cot\theta. \end{array}$$

Substitute them into Lecture 17's Ricci contraction,

$$\mathrm{Ric}_{\mu\nu} = \partial_\rho\Gamma^\rho_{\nu\mu} - \partial_\nu\Gamma^\rho_{\rho\mu} + \Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\nu\mu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\rho\mu}.$$

For example, $\Gamma^\rho_{\rho r} = 2/r$ and the tt entry is

$$\mathrm{Ric}_{tt} = \frac{(f')^2 + ff''}{2} + \frac{ff'}{r} - \frac{(f')^2}{2} = f\left(\frac{f''}{2} + \frac{f'}{r}\right).$$

The full substitution gives zero off-diagonal entries and

$$\mathrm{Ric}_{rr} = -f^{-1}\left(\frac{f''}{2} + \frac{f'}{r}\right), \qquad \mathrm{Ric}_{\theta\theta} = 1 - f - rf', \qquad \mathrm{Ric}_{\phi\phi} = \sin^2\theta\ \mathrm{Ric}_{\theta\theta}\,.$$

For $f = 1 - 2M/r$, the combinations are $f''/2 + f'/r = -2M/r^3 + 2M/r^3 = 0$ and $1 - f - rf' = 2M/r - 2M/r = 0$. Thus every Ricci component is zero. Smoothness extends this tensor equality across the angular chart poles.

The metric coefficients are independent of t and ϕ. The coordinate formula for the Lie derivative gives $\mathcal{L}_{\partial_t} g = \mathcal{L}_{\partial_\phi} g = 0$. Lecture 16's Killing conservation theorem makes $g(\partial_t, \dot{\gamma}) = f\dot{t}$ and $-g(\partial_\phi, \dot{\gamma}) = r^2\sin^2\theta\dot{\phi}$ constant. ∎

The divergence of f^{-1} at $r = 2M$ is a coordinate singularity. Indeed, put $v = t + r_*(r)$, where $r_*' = f^{-1}$. Substitution of $\mathrm{d}t = \mathrm{d}v - f^{-1}\mathrm{d}r$ gives

$$\mathrm{d}s^2 = f\,\mathrm{d}v^2 - 2\mathrm{d}v\,\mathrm{d}r - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Its determinant is $-r^4\sin^2\theta$, nonzero at $r = 2M$ away from the angular coordinate poles. These coordinates extend the metric there. In the Schwarzschild black-hole extension, this surface is the future *event horizon*: the boundary of the region from which future-directed light signals can reach arbitrarily distant observers. That global causal identification is cited, rather than inferred from a diverging coefficient; see Carroll, §7, pp. 183–185.

19.5 The massive orbit equation

Consider a future timelike Schwarzschild geodesic parametrized by proper time. Rotations of the angular coordinates are isometries because they preserve the round-sphere factor. Rotate them so the initial position and tangential velocity lie in the equatorial plane. The θ equation has the solution $\theta = \pi/2$, $\dot\theta = 0$ for these initial data. Uniqueness of the geodesic equation keeps the orbit in that plane. The constants E and L are its energy and angular momentum per unit rest mass.

Proposition 19.4 (Schwarzschild orbits · *proved*)**.** *A proper-time-parametrized equatorial timelike geodesic obeys*

$$\dot r^2 + f\left(1 + \frac{L^2}{r^2}\right) = E^2.$$

If $L \neq 0$, use ϕ as parameter and put $u = 1/r$. Then, with primes now denoting derivatives with respect to ϕ,

$$u'' + u = \frac{M}{L^2} + 3Mu^2.$$

Proof. The normalization $g(\dot\gamma, \dot\gamma) = 1$ gives

$$1 = f\dot t^2 - f^{-1}\dot r^2 - r^2\dot\phi^2 = \frac{E^2 - \dot r^2}{f} - \frac{L^2}{r^2}.$$

Rearranging gives the radial energy equation. To retain turning points, use the radial geodesic equation itself:

$$\ddot r = -\frac{ff'}{2}\dot t^2 + \frac{f'}{2f}\dot r^2 + rf\dot\phi^2 = -\frac{M}{r^2} + \frac{L^2}{r^3} - \frac{3ML^2}{r^4}.$$

The last equality substitutes the constants and the normalization just proved. It never divides by $\dot r$. Since $\dot\phi = Lu^2 \neq 0$, the chain rule gives $\dot r = -Lu'$ and $\ddot r = -L^2u^2u''$. Substitute in the radial equation and divide by $-L^2u^2$ to obtain the result. ∎

The Newtonian orbit equation lacks the term $3Mu^2$. For radial motion $L = 0$, the radial energy equation still applies, but ϕ cannot serve as parameter.

19.6 A circular orbit and its effective potential

Define the *effective potential* for fixed L by

$$U_{\mathrm{eff}}(r) = f(r)\left(1 + \frac{L^2}{r^2}\right).$$

The radial equation is $\dot r^2 + U_{\mathrm{eff}} = E^2$ and $\ddot r = -U'_{\mathrm{eff}}/2$. The figure plots an exact example of this potential.

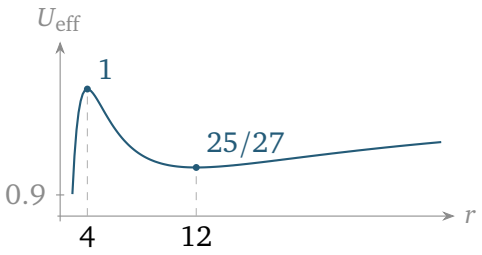


Figure 19.3: For $M = 1$ and $L = 4$, the potential has a maximum at $r = 4$ and a minimum at $r = 12$. The plot uses $2.9 \le r \le 30$ and a truncated vertical axis.

Example 19.5 (A stable circular orbit)**.** Find a massive circular orbit at radius $r_0 = 12M$. For a general circular radius $r_0 > 3M$, solving $U'_{\text{eff}}(r_0) = 0$ and $E^2 = U_{\text{eff}}(r_0)$ gives

$$L^2 = \frac{Mr_0^2}{r_0 - 3M}, \qquad E^2 = \frac{(r_0 - 2M)^2}{r_0(r_0 - 3M)}.$$

With the angular momentum held fixed, the second derivative there is

$$U''_{\text{eff}}(r_0) = \frac{2M(r_0 - 6M)}{r_0^3(r_0 - 3M)}.$$

It is positive for $r_0 > 6M$, so the radial linearization has a restoring force. At $r_0 = 12M$ the constants are $L = 4M$ and $E^2 = 25/27$. This gives the potential minimum marked in the figure after setting $M = 1$.

Self-check. Recover the inward Newtonian acceleration from Γ^r_{tt} far from the Schwarzschild source.

Answer. The table gives $\Gamma^r_{tt} = ff'/2 = M(r - 2M)/r^3 = M/r^2 + O(M^2/r^3)$. For slow motion, the geodesic equation contributes $-\Gamma^r_{tt}$ to the coordinate-time radial acceleration. Its leading value is $-M/r^2$, the attractive Newtonian result.

Summary. A weak static Lorentzian metric gives Newton's equation for slow free fall. The cited Einstein–Hilbert variation gives Einstein's equation with positive $8\pi G_N T_{\mu\nu}$ in our conventions. Contracting the second Bianchi identity gives covariant conservation. The Schwarzschild exterior has zero Ricci tensor. Its Killing constants, unit timelike normalization and radial geodesic equation give the massive orbit equation $u'' + u = M/L^2 + 3Mu^2$ in geometric units, for $L \neq 0$. Lecture 20 describes fields that require several local frames.

Reading. Nakahara, §7.10.1–2 and Example 7.15. Carroll, *Lecture Notes on General Relativity* (1997), §4, especially pp. 104–116, and §7, pp. 172–185.